ARTEFACT · AI ENGINEER TECHNICAL CASE

Empório da Música
Customer Service Agent

Candidate: Joao Jose

Career: AI Engineer – Full-Stack para Aplicações com IA Generativa

> **Public GitHub repository**
> https://github.com/Josebentivi/Desafio-T-cnico-AI-Engineer-2026

Solution

A Python prototype for the fictional music store in Campo Grande, Brazil. It provides a responsive chat, a command-line interface, persistent conversations, and a read-only operations panel. The assistant is instructed to answer in English, follow the store persona, consult tools before stating operational facts, and redirect out-of-scope requests.

Technical approach

- OpenAI Responses API with **gpt-6-luna** and typed function calls for catalog, orders, policy, shipping, and store hours.
- SQLite for operational data and conversation state; FTS5 plus embeddings and reciprocal rank fusion for hybrid policy retrieval.
- FastAPI, Jinja2, HTMX, and server-sent events for the web interface. The README explains setup, decisions, assumptions, and limitations.

Example: live catalog lookup

Customer: How much is the Yamaha C40?

Assistant: The Yamaha C40 Nylon Natural costs R$ 599,90. With PIX, it is R$ 569,91. We have 12 in stock.

Translated from a prior Portuguese-language run; the observed tools were `search_products` and `get_product`. Values reflect the supplied dataset at the time. The repository includes five documented examples.

Limitations and disclosure

The prototype requires an OpenAI API key for chat; without embeddings, policy retrieval falls back to text search. It has no authentication and is intended for local use. Actual delivery dates and a holiday calendar are not present. Customer data supplied with the challenge is included in the public repository.

OpenCode assisted with implementation, documentation, testing, and debugging. The repository documents the workflow, verification, and further limitations.